

Leveraging motion perceptibility and deep reinforcement learning for visual control of nonholonomic mobile robots

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ABSTRACT

This paper introduces a novel deep reinforcement learning framework to tackle the problem of visual servoing of nonholonomic mobile robots. The visual control of nonholonomic mobile robots becomes particularly challenging within the classical paradigm of visual servoing, mainly due to motion and visibility constraints, which makes it impossible to reach a given desired pose for certain configurations without losing essential visual information from the camera field of view. Previous work has demonstrated the effectiveness of deep reinforcement learning in addressing various vision-based robotics tasks. In light of this, we propose a framework that integrates deep recurrent policies, intrinsic motivation, and a novel auxiliary task that leverages the interaction matrix, the core of classical visual servoing approaches, to address the problem of vision-based control of nonholonomic robotic systems. Firstly, we analyze the influence of the nonholonomic constraints on control policy learning. Subsequently, we validate and evaluate our approach in both simulated and real-world environments. Our approach exhibits an emergent control behavior that enables the robot to accurately attain the desired pose while maintaining the desired visual content within the camera's field of view. The proposed method outperforms the state-of-the-art approaches, demonstrating its effectiveness, robustness, and accuracy.

1. Introduction

1.1. Motivation

In the realm of robotics, nonholonomic systems refer to a class of robots that possess motion constraints limiting their ability to reach arbitrary poses in their configuration space instantaneously. These constraints typically arise from the physical design of the robot and result in a non-integrable distribution of velocities, meaning that the robot cannot move in any direction at a given instant [1].

The visual servoing (VS) of nonholonomic mobile robots is a well-established problem in the literature [2]. VS paradigm refers to a set of methods that utilize computer vision data to control the motion of dynamic systems by minimizing an error between visual features extracted from a desired image and from images acquired during the process [3]. A control law is designed with respect to the chosen visual feature to compute the appropriate velocities to drive the robot to the desired pose. The main issue in solving VS of nonholonomic mobile robots is the motion constraints, which limit the robot's motion to a subset of the configuration space, making it challenging to reach the desired pose without losing essential visual information from the camera's field of view (FOV) (Fig. 1).

Several methodologies have been proposed to address the VS of nonholonomic mobile robots. Gans et al. [4] introduced a feedback control system that incorporates various control modes to smoothly rotate the robot, ensuring that the desired visual content remains consistently within the camera's FOV. However, this control approach is based on a finite state machine which does not capture the complex real-world settings. Other works suggested adding a pan and tilt unit onboard, to increase the camera's degrees of freedom (DOF) which would allow for maintaining the desired visual features in the camera FOV [5]. Yet, this method relies on pre-recording the images from the ground truth trajectory of the robot. Zhang et al. introduced a novel motion estimation technique that leverages the continuity property of robot's movements to enhance robustness against FOV loss problems [6]. Omnidirectional cameras also have been shown to alleviate the FOV problem and guarantee the visibility of desired visual features [7]. However, these methods [6,7] face the challenge of detecting, matching, and tracking geometric features. This challenge can be overcome by using photometric VS. The extension of photometric VS from perspective to omnidirectional vision sensors for nonholonomic platforms was suggested in [8], but it comes with the drawback of a limited convergence domain.

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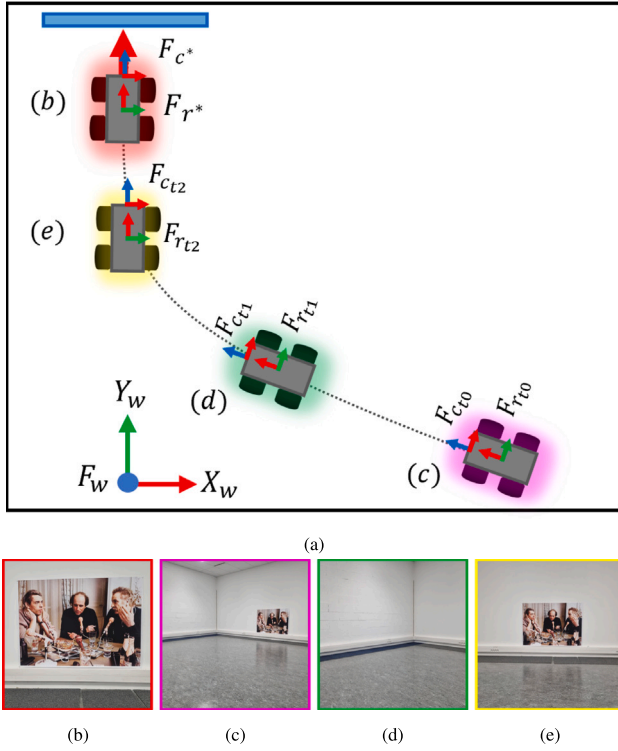


Fig. 1. Test case where classical VS methods fail due to visibility/motion constraints: (a) An example of a trajectory for the robot to reach the desired pose, (b) The desired image captured at the desired pose, (c), (d), and (e) are the images acquired by the robot at poses (c), (d), and (e), respectively.

In parallel, deep reinforcement learning (DRL) has emerged as a promising framework to address the challenges associated with vision-based robotics such as autonomous driving [9], multi-robot navigation [10], and manipulation and grasping [11]. Although the problem of learning VS for robotic manipulators has been investigated in prior research, leveraging DRL to address the VS of nonholonomic mobile robots remains largely unexplored. Thus, in this work, we investigate whether DRL is capable of learning a control policy that can make a nonholonomic robot with a limited FOV autonomously and precisely converge to a desired pose simply represented by an image. We argue that, unlike classical VS methods, DRL goes beyond feature extraction and camera modeling, which are often central to classical VS approaches. More specifically, DRL enables the emergence of control strategies that might be impossible to achieve or would require complex analytical models. Beyond that, studying such tasks allows for better understanding of the synergy between DRL algorithms and nonholonomic robotic platforms.

1.2. Related work

This section provides a comprehensive review of existing research related to the addressed problem, specifically focusing on three key domains: VS of nonholonomic mobile robots, DRL for VS of robotic manipulators and target-driven control. These areas of study share a common objective of controlling a robot using visual information extracted from current and desired images.

VS of nonholonomic mobile robots: Existing VS approaches can be divided into two categories: image-based (IBVS) and position-based (PBVS). IBVS uses visual features like points, lines or pixels intensities to compute errors between current and desired images and a control law to compute the robot's velocities in order to reduce this error [3]. However, IBVS suffers from issues such as image singularities, local

minima, and challenges associated with depth estimation, particularly without a known 3D model [12,13]. PBVS, on the other hand, estimates the 3D pose between configurations using image features, but its reliance on prior 3D models and camera parameters reduces its feasibility in real-world scenarios [14].

In practice, addressing the VS of a nonholonomic mobile robot using IBVS or PBVS is not sufficient due to the limited FOV of the camera. Once features are lost, the VS task cannot be carried out as expected. Thus, it is important to consider visibility constraints and the nonholonomic constraints when designing the servo controller. In this context, several methods have been proposed to address the loss of features from the camera's FOV, referred to as hybrid VS (HVS). These methods often integrate IBVS and PBVS approaches along with other techniques, such as model predictive control (MPC), fuzzy logic methods or RL [15–17]. In general, HVS methods can be separated into two categories. The first category aims to avoid FOV loss by optimizing the trajectory of features in the image frame using MPC methods, demonstrating strong robustness due to long-horizon trajectory optimization or planning [18,19]. These methods suffer from a heavy computational burden, which may limit their application, especially in real-time tasks as argued Heshmati et al. [20]. The second category of methods aims at ensuring the visibility of the desired visual content at the next state. Chesi et al. [15] proposed a switching controller that relies on a PBVS strategy to keep visual features within the camera's FOV based on their position in the image frame. Similarly, potential function-based control was used to guarantee that the features remain in the FOV [21,22]. However, these methods require careful tuning, such as setting the conditions in the first method and determining the size and shape of the potential function in the latter. While the previous methods rely on heuristics to avoid the FOV loss problem, the rise of RL and DRL has enabled their combination with IBVS and PBVS to autonomously learn strategies that keep features within the camera's FOV. Wang et al. [17] and Miljkovic et al. [23] applied Q-learning in the context of VS under constrained FOV by setting a switching rule between the Q-learning agent and the VS controllers. Building on this work, a deep Q-network (DQN) was coupled with a fuzzy controller for learning rate tuning [24], or to dynamically adjust the gain of an IBVS controller [25]. Similarly, the same ideas were adopted for nonholonomic mobile robots under constrained FOV, where Jin et al. [26] used a deterministic policy gradient algorithm (DDPG) to adjust the gains of an IBVS controller to keep the image features in the FOV, arguing that using DRL with a continuous action space yields better results. Likewise, inspired by Chesi et al. [15], Wu et al. [27] combined a DQN and an MPC controller using a heuristic switching rule based on the position of the features in the image frame to switch between these controllers. The works most closely related to ours those proposed by Li et al. [28] and Choi et al. [29], both attempt to address the challenge of FOV loss and learning VS of mobile robots using end-to-end DRL. Li et al. [28] utilized dense image correspondence as input features to train a DQN with a novel-shaped reward function. In contrast, Choi et al. [29] proposed incorporating keypoint features from both desired and current images as additional inputs for learning a control policy.

DRL for VS of robotic manipulators: Several works have been proposed to learn VS of six-DOF robotic manipulators using DRL approaches. A learned deep end-to-end control policy for VS of six-DOF robotic manipulators using a deep convolutional neural network (CNN) has been introduced [30]. While the proposed approach demonstrated the potential of DRL in solving the task, the precision at convergence was poor compared to classic VS methods. Sadeghi et al. [31] introduced a method that combines Monte Carlo estimation with recurrent policies, focusing on the problem of reaching semantic targets. Joshi et al. [11] used images from a multi-view camera setup as input for learning a VS control policy. Their study demonstrated that the multi-view approach enhances the accuracy of the VS task compared

to classic single-view approaches. However, the multi-view camera setup presents limited practical applications. Asayesh et al. [32] proposed a novel approach integrating CNNs, DRL, and optimal control to tackle VS challenges for robotic manipulators. Their work highlights the advantages of DRL-based methods over traditional visual servoing techniques. Since DRL approaches require massive amounts of data, the training stage is often conducted in simulations, which introduces limitations when deploying the trained policy on a real robot [33]. This issue was addressed by Pederson et al. [34], who proposed using a generative adversarial network (GAN) to translate the real images acquired during an experiment to the virtual domain of the synthetic training data, which requires pretraining the GAN on synthetic data prior to training the control policy.

Target-driven control: This section encompasses related works, outlining methods drawn from ImageGoal navigation, and visual active tracking (VAT), respectively. Closely related to the VS of nonholonomic mobile robots is ImageGoal navigation. This problem can be traced back to Zhu et al. [35], who proposed a framework that incorporates a deep Siamese encoder and an asynchronous actor-critic agent to solve a navigation task. An approach that combines deep recurrent policies and a novel auxiliary task for observation reconstruction was presented by Kulhanek et al. [36]. Imitation learning [37], as demonstrated in previous research [38], has proven its efficiency in ImageGoal navigation tasks. The paper [38] introduced a DRL model that leverages the advantages of variational autoencoders for observation prediction, trained with imitation learning, along with auxiliary tasks for collision avoidance. This requires training the generative model used in this method prior to the control policy. The problem of transferring navigation policies from simulation to real environments was highlighted by Sadeghi et al. [39], who showed that using visual semantics to train a recurrent CNN, allows for one-shot transfer of the learned policies to the real robots. Mezghani et al. [40] introduced an approach based on the use of an attention model to select information for the agent from episodic memory. Majumadar et al. [41] proposed a method that uses large-scale pre-training and image-text alignment models to enable zero-shot object-goal navigation by projecting language and visual goals into a shared multimodal semantic embedding space. Yadav et al. [42] built upon their previous work [43,44] and the previously introduced methods, proposing an ImageGoal navigation method based on a vision transformer (ViT) as an encoder for the target and current images, achieving SOTA results in ImageGoal navigation on the HM3D dataset [45]. Closely related to the ImageGoal navigation task is the VAT task, which focuses on continuous observation or interaction with a target object within the robot's FOV. Various studies have approached this task using DRL methods. For example, Luo et al. [46] trains agents in simulated environments using domain randomization for better generalization. In contrast, Zhong et al. [47] employs an asymmetric dueling training process with an adversarial target to develop effective policies. Additionally, [48] enhances performance by leveraging continuous actions and auxiliary task learning in an end-to-end architecture. Finally, Dionigi et al. [49] removes the initial assumption that the target is within view, allowing agents to explore unknown environments to locate and track it. While the VAT task is highly relevant to VS and ImageGoal tasks, its main objective differs, as it focuses on ensuring continuous observation or interaction with either a dynamic object or a static object in the scene.

Discussion: Despite progress in the aforementioned methods, addressing the VS task for nonholonomic mobile robots using end-to-end approaches remains largely unexplored. Existing DRL approaches for VS tasks have primarily focused on robotic manipulators, often overlooking nonholonomic mobile robots, which are considered more complex in the VS literature [50]. In contrast, existing methods that address the VS of nonholonomic mobile robots are primarily framed within the traditional VS framework, i.e., a modular feedback control closed loop that includes feature extraction, matching and tracking, camera

models, and the construction of an interaction matrix to design a control law. We argue that employing an end-to-end DRL approach enables the VS task to leverage information from entire images, which are directly mapped to control velocities for robot control, thereby avoiding modularity, where the failure of a single block can potentially lead to system failure. Furthermore, DRL-based control policies enable the emergence of solutions that might be infeasible or require highly complex analytical models. The previously presented SOTA methods for the VS of nonholonomic mobile robots incorporate DRL, however they do not use it in for end-to-end control and instead integrate it into the classical VS pipeline. In addition, these methods combine various techniques, leading to high computational costs and difficulties in implementation and system tuning. On the other hand, ImageGoal navigation methods tend to assume that the desired image is reached within a 1 m distance from the desired position, in contrast to VS, where the goal is to accurately position the robot. Furthermore, ImageGoal navigation methods train the robot to operate in a discrete action space, i.e., using constant control velocities, which may result in unstable behavior upon convergence. Additionally, the training is typically conducted in grid-world-like environments.

1.3. Contributions

In this paper, we propose to combine auxiliary tasks with intrinsic motivation [51] in training a nonholonomic mobile robot. Intrinsic motivation defines a set of methods based on assigning an exploration bonus to reward the agent for improving its knowledge about the environment or visiting new states. This bonus, also known as intrinsic rewards, is generated by the agent itself in a self-supervised manner, independently of its environment. In a vision-based control context, intrinsic motivation allows the robot to efficiently explore the space state during the training phase. Auxiliary tasks are additional biases that are optimized in addition to classic DRL objectives that encourage the robot to learn features relevant to the surrounding environment. In this light, we introduce a novel DRL framework that incorporates deep recurrent policies, intrinsic motivation, and a novel auxiliary task for learning VS of nonholonomic mobile robots. Our proposed auxiliary task leverages the interaction matrix, the core of classical VS literature, and extrapolates this notion to enhance DRL-based VS methods.

Firstly, we highlight and discuss the difficulties faced by a nonholonomic robot in efficiently exploring the environment due to its motion constraints, and consequently, the challenges associated with learning complex trajectories. We then evaluate our method in a simulated environment, demonstrating its ability to solve the task. Furthermore, we validate its transferability and usability on a real robot.

Our contributions can be summarized as follows:

- Demonstration of the effects of the nonholonomic constraints on control policy learning.
- A novel auxiliary task for learning vision-based control.
- A novel DRL method that combines deep recurrent policies, intrinsic motivation and auxiliary tasks.
- Comparisons with the state of the art (SOTA) methods, as well as evaluations of the advantages and limitations of using DRL in solving VS of nonholonomic mobile robots.

1.4. Outline

The rest of this paper is organized as follows: Section 2 delineates a systematic formulation of the problem at hand. In Section 3, the proposed approach is presented, showcasing the different modules of our DRL model. Section 4 describes the experimental protocol and training details. Subsequently, Section 5 presents simulation results, provides a comparative analysis, and presents results from experiments in the real world. Lastly, Section 6 offers conclusive insights, highlights the limitations encountered, and outlines potential directions for future research.

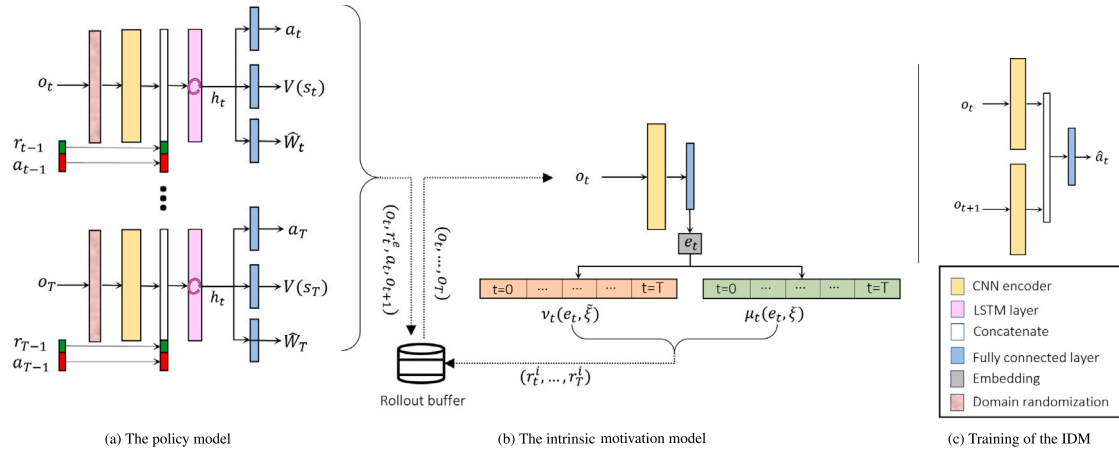


Fig. 2. The proposed DRL framework: (a) Illustration of the architecture of our policy model, (b) The intrinsic motivation model responsible for generating intrinsic rewards, (c) Training of the IDM model.

2. DRL for VS of nonholonomic mobile robots

This section presents a formulation of learning VS for nonholonomic mobile robots. This involves articulating a systematic framework to address the VS task using DRL approaches.

2.1. Problem statement

Consider a scenario in which a nonholonomic mobile robot operates with the aim of reaching a desired pose, denoted as $p^* = (x^*, y^*, \beta^*)$, defined by an image I^* , where (x^*, y^*) is the desired position and β^* is the desired orientation expressed in a fixed world reference coordinates frame (Fig. 1). Traditional VS methods commonly rely on extracting visual features from the desired image I^* , and from current image I_t captured at $p_t = (x_t, y_t, \beta_t)$ at instant t . Here, (x_t, y_t) denotes the instantaneous position of the robot, and β_t is its orientation. A control law is designed to minimize the error between the desired and current visual features using the interaction matrix that expresses the displacement of the visual features with respect to the motion of the robot. In contrast, DRL methods present a compelling paradigm that enables learning a representation capable of directly mapping the desired and input images to control velocities, thus, alleviating the need to analytically model complex control strategies with respect to the perceived visual features.

2.2. Problem formulation

We propose to learn a goal-conditioned stochastic policy function denoted as π using a deep neural network as a nonlinear function approximator. Since images do not explicitly convey information about the robot's state, VS tasks can be modeled as a partially observable Markov decision process (POMDP) [52]. In reinforcement learning (RL), a POMDP model is defined by the tuple (S, A, O, P, R, γ) , where S is the state space that represents the pose of the robot p . The set $P(s' | s, a)$ is the transition function, denoting the likelihood of transitioning to state s' when action a is executed in state s . $\gamma \in [0, 1]$ denotes the discount factor. The action space, observation space, and reward set are denoted by A , O , and R respectively, and will be comprehensively detailed in subsequent sections.

The DRL agent is initially placed in a state denoted as s_0 , where the objective is to learn the optimal sequence of actions (control velocities) to minimize the error between the current and the desired images. During the training process, the robot perceives an observation denoted as o_t , executes an action denoted as a_t , and receives a deterministic reward denoted as $r_t(s_t, a_t)$. At each timestep t , the action a_t is sampled

according to the equation:

$$a_t \sim \pi_\theta(o_t, a_{t-1}, r_{t-1}), \quad (1)$$

where a_{t-1} and r_{t-1} denote the executed action and the perceived reward at instant $t-1$ respectively. θ represents the parameters of the neural network utilized in the policy function. The policy is trained to maximize the expected discounted return denoted by $R_t = \mathbb{E}_\pi \left[\sum_{k=0}^T \gamma^k r_{t+k+1} \right]$, where T is the episode length.

2.3. Learning setup

2.3.1. Observation space

The observations in our setting consist of a pair of RGB images, denoted $o_t = \{I_t; I^*\}$, where I_t represents the current image acquired at time t , and I^* represents the desired image. The desired image, captured at pose p^* , and the current image captured at p_t are obtained using the RGB camera mounted on the robot.

2.3.2. Action space

Learning real-world robotics applications requires consideration of low-level mechanical aspects. Consequently, the adoption of a physical engine becomes imperative to effectively account for the underlying physics involved in such low-level control. Additionally, tasks that involve precise positioning require high control sensitivity. In light of these requirements, we train our model using a physical simulator, focusing on the continuous control of linear and angular velocities expressed in the robot frame (Fig. 1) and respectively denoted as v_x and v_β . At each timestep t , the robot executes an action, denoted as $a_t = (v_{x_t}, v_{\beta_t})$.

2.3.3. Reward function

At each timestep t , the agent receives a reward $r_t = r_t^e + \eta r_t^i$, which is the sum of two distinct components: an extrinsic reward r_t^e that relates to the task at hand and, an intrinsic reward r_t^i that originates from the intrinsic motivation mechanism within the agent. Here, η is a variable weighting parameter for the relative importance of the intrinsic reward compared to the task reward. This section focuses solely on extrinsic reward. The generation of the intrinsic reward is detailed in Section 3.2.

Our VS task reward function is defined as the Euclidean distance between the current pose and the desired pose of the robot:

$$r_t^e = \lambda_d * \exp \left(-K \sqrt{(x_t - x^*)^2 + (y_t - y^*)^2} \right) + \lambda_\beta * \exp \left(-K \sqrt{(\beta_t - \beta^*)^2} \right). \quad (2)$$

In this formulation, the function $\exp(x) = e^x$ is employed to map reward values within the range of 0 to 1. This mapping serves to standardize the rewards as input for the neural network. The weighting coefficients λ_d and λ_p are utilized to adjust the relative importance of the robot's position and orientation. Additionally, the parameter K is introduced to control the sparsity of the reward signal.

3. Learning VS for nonholonomic mobile robots

This section introduces our proposed DRL framework (Fig. 2), which combines two modules: a policy model and an intrinsic motivation model. The policy model aims to learn the mapping from input images to control velocities, while the intrinsic motivation model is designed to generate intrinsic rewards to enhance the agent's exploration during training. Each component of the modules is described hereafter.

3.1. Policy model

3.1.1. Domain randomization

In our proposed policy model (Fig. 2(a)), we first apply visual domain randomization to the input images to facilitate the transfer of our control policy from simulation to the real world. We employ classic data augmentation techniques to apply random transformations to the input images. Specifically, variations in brightness, contrast, saturation, and hue levels are introduced to the input images in order to capture invariant features of the scene (Fig. 3). Additionally, two types of Gaussian noise are incorporated. The first is uncorrelated noise, sampled at the beginning of each training episode, while the second is correlated noise, sampled at each timestep t . The inclusion of Gaussian noise is intended to mimic the noise present in the real cameras [53].



Fig. 3. Illustration of visual domain randomization applied to the input images.

3.1.2. Actor-critic model

Firstly, the resulting images from domain randomization are passed through a CNN encoder to extract essential visual features. The extracted features from the current and desired images are then concatenated with the actions and rewards from instant $t - 1$ to form a joint representation with the observation. The joint representation is then passed through a long-short-term memory layer (LSTM) [54]. The LSTM layer allows the learning of temporal representations between observations, which accounts for the partial observability in VS [55]. Finally, the model outputs the robot's control velocities $a_t = (v_{x_t}, v_{\theta_t})$, an estimate of the value function (V_f) $V(s_t)$. Indeed, an actor-critic approach combines both policy-based and value-based methods, allowing for simultaneous policy optimization and V_f estimation.

To train our actor-critic model, we use a continuous Proximal Policy Optimization (PPO) actor-critic style architecture augmented with the Generalized Advantage Estimation (GAE) technique, similar to the methodology described in [56]. This choice is justified by the algorithm's stability throughout the training process, its ability to effectively parallelize multiple environments, and its robustness to

training hyperparameters. In addition, given the on-policy nature of PPO, we avoid using off-policy continuous control algorithms, as using a replay buffer with our observation space requires large computational resources. The incorporation of GAE enables improved estimation of the advantages associated with state-action pairs, which contributes to improving the sample efficiency and stability of training. In this setting, V_f is estimated by minimizing a mean squared error (MSE) loss function, denoted as $\mathcal{L}^{V_f}$. PPO operates as a policy gradient RL algorithm that alternates between sampling data and optimizing a surrogate loss function in a cyclic manner.

The loss function related to the PPO algorithm can be expressed as follows:

$$\mathcal{L}^{PPO} = \frac{1}{|\mathcal{M}|T} \sum_{\tau \in \mathcal{M}} \sum_{t=0}^{T-1} \left[\min \left(R_t(\theta) \hat{A}_t, \zeta(R_t(\theta), \delta) \hat{A}_t \right) \right], \quad (3)$$

where $\mathcal{M}$ is the set of the sampled trajectories $\tau = (o_t, a_t, r_t, \dots, o_T, a_T, r_T)$. $\hat{A}_t$ represents an estimate of the advantage function, which is defined as:

$$A_t(s_t, a_t) = Q(a_t, s_t) - V(s_t). \quad (4)$$

The state-action value function is denoted as:

$$Q(s_t, a_t) = \mathbb{E}_{a_{t'} \sim \pi_\theta} \left[\sum_{t'=t}^T \gamma^{t'-t} r_{t'} \mid s_t, a_t \right]. \quad (5)$$

The value function is expressed as:

$$V(s_t) = \mathbb{E}_{a_{t'} \sim \pi_\theta} \left[\sum_{t'=t}^T \gamma^{t'-t} r_{t'} \mid s_t \right]. \quad (6)$$

The term denoted as $R_t(\theta) = \frac{\pi_\theta(a_t | s_t)}{\pi_{\theta_{old}}(a_t | s_t)}$ represents the ratio of probabilities between the current policy $\pi_\theta(a_t | s_t)$ and the old policy $\pi_{\theta_{old}}(a_t | s_t)$. To ensure stability during training, a clipping function, often denoted $\zeta(\theta, \delta) = \text{clip}(R_t(\theta), 1 - \delta, 1 + \delta)$, is employed with a threshold δ .

3.1.3. Motion perceptibility as an auxiliary task

In DRL, a common approach to improve the main task's performance is to train the agent to perform additional auxiliary tasks. Although it is not yet clear how relevant auxiliary tasks improve the agent's representation [57], by learning to perform these tasks, the agent can acquire useful information that can be leveraged to improve performance on the task in question. In light of this, we propose a novel auxiliary task, formulated as a regression problem, specially designed for VS learning. As previously mentioned, the core of usual VS approaches is the interaction matrix L that expresses the relation between the time variation of the visual features considered with respect to the displacement of the camera [3]. In our case, the interaction matrix is not used since the agent has to autonomously learn a control policy based on its observations. Nevertheless, incorporating this information into the DRL process would be beneficial. In their research on active vision [58], Hutchinson et al. introduced the concept of a vision system's ability to perceive the movement of a scene based on the visual content captured in its FOV. This concept, called motion perceptibility (MoP) is a global quantitative measure that can be computed from the interaction matrix following:

$$W = \sqrt{L^\dagger L}, \quad (7)$$

where $\cdot^\dagger$ denotes the Moore-Penrose inverse. Classical VS frameworks therefore exploit this measure intrinsically in the control law. We assume that integrating this notion into learning VS could be beneficial in indirectly providing the agent with a relationship between its actions and observations, i.e., between its visual perception and its movements, thereby assisting it in making improved control decisions. Thus, we introduce the MoP auxiliary task, a novel auxiliary task where an agent learns to predict MoP values from its observations. To compute the ground truth W values, we use the information contained in the current image acquired by the robot to construct the photometric interaction matrix [59], which is then used to compute the MoP.

The actor-critic model is equipped with an additional prediction head that takes as input the contextual representation h_t produced by the LSTM layer (Fig. 2(a)) and outputs a prediction of our auxiliary task. The proposed auxiliary task approximates a function $f_\rho(h_t) = \hat{W}_t$ that predicts MoP values where ρ denotes the parameters of the learned function. MoP auxiliary task loss is given by:

$$\mathcal{L}^{MoP} = \frac{1}{|\mathcal{M}|T} \sum_{\tau \in \mathcal{M}} \sum_{t=0}^{T-1} \left[(\hat{W}_t - W_t)^2 \right], \quad (8)$$

where W_t is the ground-truth MoP value. It is important to note that this additional prediction task is used only during training time to encourage the robot to learn the dependency between the executed control velocities and the changes in visual inputs.

Finally, the total loss of our whole policy model is the sum of the PPO loss, the value function loss, the entropy regularization term and the auxiliary task loss:

$$\mathcal{L}^{tot} = \mathcal{L}^{PPO} + \lambda_{V_f} \mathcal{L}^{V_f} - \lambda_H H(\pi_\theta(s)) + \lambda_{MoP} \mathcal{L}^{MoP} \quad (9)$$

Here, $H(\pi_\theta(s))$ is an entropy regularization term which aims at enhancing the exploration of the policy. λ_{V_f} , λ_{MoP} and λ_H weight the relative importance of the value function, MoP auxiliary task, and the entropy regularization term, respectively. At inference time, i.e., when deploying the model on the robot, V_f and MoP predictions are not considered. Note that only the output of the actor is taken into account to select the appropriate control velocities.

3.2. Intrinsic motivation model

As shown in our experimentation (Section 5.1) the nonholonomic constraints constitute a considerable challenge to the exploration capabilities of vanilla DRL agents within the environment. To overcome this problem, we propose using an exploration bonus based on an approximation of the Rényi divergence introduced in [60]. Our motivation for employing this particular method stems from its ability to incorporate observations from both old and current policies when computing intrinsic rewards. We postulate this would allow the robot to explore a more diverse set of trajectories during training. This model (Fig. 2(b)) is used only during training to generate intrinsic rewards. Let $p_\pi(s)$ represent the state distribution induced by the policy that has been learned, $p_{\bar{\pi}}(s)$ denote the state distribution resulting from the previous policy. α is the order of the divergence where $\alpha \in [0, \infty]$ and $\alpha \neq 1$. The Rényi divergence for the policy π can be defined as follows:

$$D_\alpha(p_\pi(s) \parallel p_{\bar{\pi}}(s)) = \frac{1}{\alpha - 1} \cdot \log \int_{\mathcal{S}} p_\pi^\alpha(s) p_{\bar{\pi}}^{1-\alpha}(s) ds. \quad (10)$$

Consider two sets of visited states from two consecutive episodes, denoted as $\xi = \{s_t\}_{t=1}^T$ and $\tilde{\xi} = \{\tilde{s}_t\}_{t=1}^T$. The approximation of the Rényi divergence is given by the following expression:

$$\begin{aligned} \hat{D}_\alpha^{k,T}(p_\pi(s) \parallel p_{\bar{\pi}}(s)) &= \\ \frac{1}{\alpha - 1} \log \left(\frac{1}{T} \sum_{t=1}^T \left[\frac{(T-1)\mu_k(e_t, \xi)}{T v_k(e_t, \tilde{\xi})} \right]^{1-\alpha} C_{k,\alpha} \right), & \quad (11) \\ \propto \frac{1}{T} \sum_{t=1}^T \left(\frac{\mu_k(e_t, \xi)}{v_k(e_t, \tilde{\xi})} \right)^{1-\alpha} \end{aligned}$$

where μ_k denotes the Euclidean distance between e_t and its k th nearest neighbor in the set ξ . v_k corresponds to the Euclidean distance between e_t and its k th nearest neighbor in the set $\tilde{\xi}$. Here, $e_t = f_\phi(o_t)$ refers to the embedding vector generated by an IDM denoted as f_ϕ . Furthermore, $C_{k,\alpha}$ is a constant given by $C_{k,\alpha} = \frac{\Gamma(k)^2}{\Gamma(k-\alpha+1)\Gamma(k+\alpha-1)}$, which is defined for $k > |\alpha - 1|$, where $\Gamma(\cdot)$ represents the Gamma function. Lastly, ϵ signifies a fixed small constant. The intrinsic reward is obtained following:

$$r_t^i(o_t) = L(\xi) \left(\frac{\mu_k(e_t, \xi)}{v_k(e_t, \tilde{\xi}) + \epsilon} \right)^{1-\alpha}. \quad (12)$$

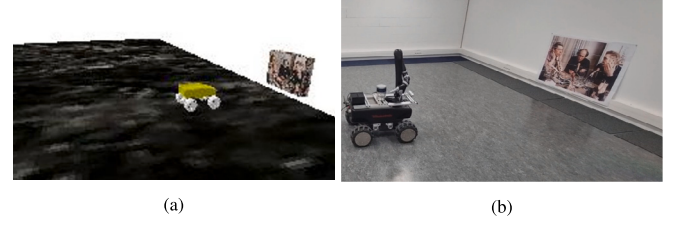


Fig. 4. View of (a) the synthetic scene used for training and (b) the real experimental room. We use a Robotnik Summit-XL nonholonomic mobile robot equipped with a front-mounted RGB camera. Note that the robotic arm is not utilized in this work.

Here, $L(\xi) = \tanh\left(\frac{1}{T} \sum_{t=0}^T \mu_1(e_t, \xi)\right)$ is a scaling coefficient which aims to avoid the robot lingering in a small region and not exploring the environment [60].

The choice of an appropriate encoder for intrinsic motivation models in DRL plays an important role in effectively representing the state for intrinsic reward computation, thereby impacting the overall performance of the agent. In this regard, we choose to employ an IDM, which has demonstrated its superiority over alternative encoders, as evidenced by the ablations presented by Burda et al. [61]. The IDM encoder is a neural network that is trained to predict the actions $\hat{a}_t$ of a given transition $f_\phi(o_t, o_{t+1}) = \hat{a}_t$. The encoder parameters, denoted as ϕ , are optimized using MSE loss (Fig. 2(c)). The fundamental idea behind this encoder is its ability to capture environmental features that directly correlate with the robot's immediate control velocities. Consequently, the intrinsic rewards received by the robot remain unaffected by variations in the environment, such as noise or alterations in visual inputs.

4. Experimental setup

This section covers the implementation details of our approach, outlines the training setup we used, and describes the evaluation protocol for comparing methods.

4.1. Environment

Models training was conducted using the Pybullet simulator [62]. The environment was designed to broadly resemble our real-world scene, featuring an empty room with no objects except a poster (Fig. 4). This particular choice served three primary purposes: (i) it is straightforward to design such environments in simulators, (ii) it enables zero-shot transfer to the real robot by making the DRL model learn global discriminate visual features, and (iii) it prevents the DRL agent from using features other than those from the desired image. It is important to note that the visual simplicity of this environment renders the task very challenging to solve via DRL. An optimal policy should be able to drive the robot to the desired pose by taking the shortest trajectory while maintaining the desired visual content in the FOV of the camera. On the other hand, our motivation for conducting a zero-shot transfer is to evaluate whether the robot can replicate the learned behavior from simulation in the real-world scene, as we deem the learned behavior to be complex.

Throughout the training, the desired pose is fixed at $p^* = (0.0 \text{ m}, 8.0 \text{ m}, 0.0^\circ)$, which corresponds to positioning the robot facing the poster at a distance of 2.0 m (Fig. 4). The robot's initial positions are then randomly sampled to ensure that the initial distance error falls within the range of 0.5 to 4.0 m. Additionally, the robot's initial orientations are randomly initialized in such a manner that the desired image remains, even partially, within the camera FOV. We further note that the design of this environment allows for a fair comparison with classical VS approaches.

Table 1
Summary of training hyperparameters.

Environment	λ_d	0.8
	λ_β	0.2
	K	0.5
	image size	80×80
	Episode length T	1500
PPO	n of environments	48
	batch size	4096
	learning rate	0.0001
	optimizer	Adam
	epochs	10
	Aux task coeff λ_{MoP}	0.2
	V_f coeff λ_{V_f}	0.5
	Entropy coeff λ_H	0.007
	GAE λ	0.95
	clip range δ	0.1
Intrinsic motivation model	discount factor γ	0.99
	ϵ	0.0001
	k	3
	η	0.1
	α	0.5
	η rate	0.0001

4.2. Training

The experiments were carried out on hardware comprising an NVIDIA RTX 3090 GPU and an Intel i9-12900KF CPU. For model training, we used the PPO agent with GAE from the stable-baselines3 library [63]. The deep recurrent actor-critic policy was trained using the training protocol detailed in Section 3.1.3, involving 48 parallel instances of our environment, each initialized randomly. Completion of one million training steps requires approximately two hours.

PPO and policy hyperparameters (Table 1) are tuned using the Tree-structured Parzen Estimator (TPE) from Optuna framework for hyperparameter tuning [64]. The weighting parameters in the loss and reward function were set empirically. λ_{MoP} was set to bias the policy training towards the PPO loss rather than the auxiliary task's loss. The choice of the parameters λ_d , λ_β and K of the reward function was made to ensure a dense distribution of rewards within the training environment. The hyperparameters of the intrinsic motivation model are the same as those used in [60]. We train our model for five random seeds. The reported results are averaged across these seeds.

4.3. Evaluation

For evaluation, we consider a pre-sampled test set composed of 100 initial poses. We perform both qualitative and quantitative evaluations of the compared methods. For qualitative evaluation, we analyze the trajectories performed on this test set. For quantitative evaluation, we consider the position error at convergence (ΔM), and the orientation error at convergence ($\Delta\beta$). We report the average error across the entire test set.

5. Experimental results

This section highlights the impacts of nonholonomic constraints on control policy learning, presents simulation and real-world results, and provides a comparative analysis.

5.1. Impact of nonholonomic constraints on learning VS

To understand the impact of the nonholonomic constraints on DRL-based VS methods, we trained a nonholonomic and an omnidirectional robot using our base agent, a simple recurrent PPO, and we then analyze their performed trajectories on our test set. For visibility purposes, we highlight the performed trajectories on a representative subset from the test set (Fig. 5).

The desired pose is represented by the red arrow. The initial orientations and the orientations at convergence are represented with arrows for each corresponding robot. Our results obviously show that the nonholonomic robot has difficulty reaching the desired pose compared to the omnidirectional one. This discrepancy arises jointly from both the nature of exploration in DRL algorithms and the motion constraints present in nonholonomic robots. More precisely, consider a scenario in which the robot needs to execute a precise sequence of T control velocities at poses $p_0, \dots, p_{T-1}$ to achieve a desired pose. The probability of achieving the desired pose can be expressed as $P(p^*) = P(a_0 | p_0)P(a_1 | p_1) \dots P(a_{T-1} | p_{T-1})$, where $P(p^*)$ is thus exponentially lower than the individual input probabilities. The omnidirectional robot, with the ability to sample control velocities along its three DOFs, can easily learn a near-optimal policy through sampling random actions. On the other hand, the nonholonomic robot, restricted to only two DOFs, faces the challenge of randomly sampling precise sequences of angular and linear velocities to explore laterally, making learning complex trajectories very difficult. To further understand this behavior, we observe the states visited by both robots during an early episode in the training stage. The heatmaps shown in Figs. 6(a) and 6(b) illustrate the distribution of states explored by the robots.

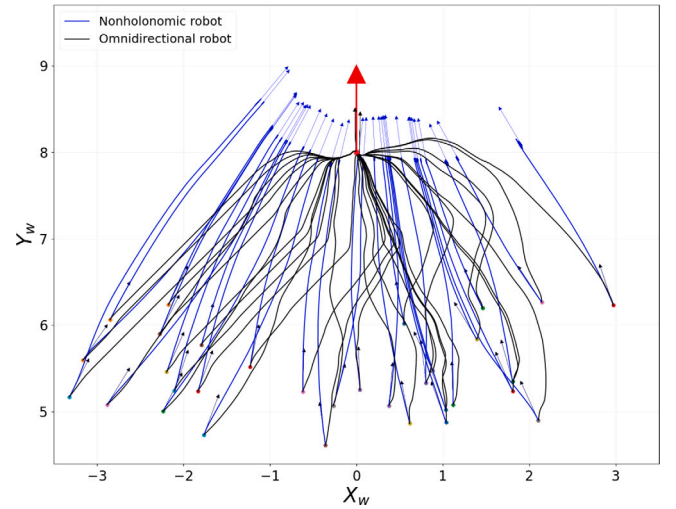


Fig. 5. Illustration of the trajectories performed by the omnidirectional and nonholonomic robots.

Both robots start from the same initial pose, $p_0 = (0.0 \text{ m}, 7.0 \text{ m}, 0.0^\circ)$. The nonholonomic robot tends to explore along straight lines due to its inability to move sideways. In contrast, the omnidirectional robot efficiently covers a large area of its local state space.

The results of this experiment suggest that the nonholonomic constraints limit the robot's capacity to leverage its local information to learn an optimal policy when trained with vanilla DRL algorithms.

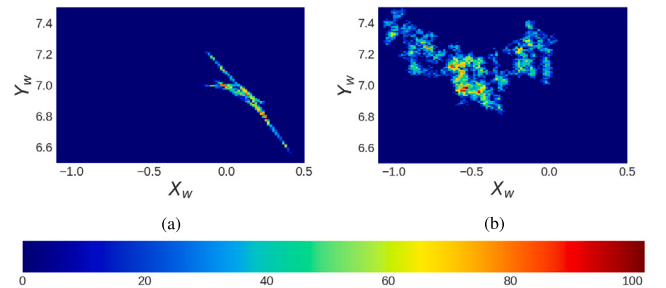


Fig. 6. Heatmaps of the visited states (position) for (a) the nonholonomic and (b) omnidirectional robots.

Hence, the need for more sophisticated exploration mechanisms, such as intrinsic motivation (Section 3.2).

5.2. Evaluation and comparison

This section has two objectives: firstly, to evaluate the benefits of each of the proposed modules included in our framework (Section 3), and secondly, to compare the performance of our method with other approaches. More specifically, we analyzed the trajectories performed by the robot on the test set after being trained using the following methods:

- **Random policy** is the simplest heuristic for learning a VS task. The robot selects random control velocities at each step.
- **IBVS** is a classical IBVS using point features from input images. To eliminate the influence of point extraction, tracking, and matching, we use the true projection of the corners of the poster (Fig. 4(a)) as visual features. The comparison with IBVS is intended for the illustration of the contrast between classic VS approaches and DRL-based approaches, and is by no means a fair comparison
- **IBVS-H** is a SOTA HVS method proposed in [26]. This method combines classic IBVS with a DDPG agent to dynamically adjust the VS gain, along with clamping linear and angular control velocities.
- **VS-ViT** is the SOTA ImageGoal navigation method approach proposed in [42]. This method is based on using a ViT encoder with a recurrent policy to train a PPO agent.
- **LVS** is the SOTA DRL approach proposed in [28]. This method is based on using dense correspondence between the current and desired images as an input for a DQN.
- **RPPO** is our base agent, it is a recurrent PPO with neither an intrinsic motivation model nor an auxiliary task. We use RPPO to train both a nonholonomic robot denoted as RPPO and an omnidirectional robot denoted as RPPO_O.
- **RPPO+IDM (ours)** is our recurrent base agent combined with the intrinsic motivation model described in Section 3.2 that uses an IDM as an encoder to embed observations.
- **RPPO+IDM+MoP (ours)** is our full model that combines our base agent RPPO, the intrinsic motivation model and the MoP auxiliary task described in Section 3.1.3.

It is important to note that all DRL-based methods are trained following the experimental protocol introduced in Section 4.2.

Fig. 7(a) shows the trajectories performed by the robot on the test set after being trained respectively with RPPO, RPPO+IDM and with the whole proposed framework RPPO+IDM+MoP. Interestingly, on the one hand, the incorporation of the intrinsic motivation model in our method (RPPO+IDM) allows the emergence of an interesting control strategy during the training. Indeed, the robot has learned to maneuver until it reaches the desired pose while maintaining the desired image in the camera's FOV, allowing it to recover from poses that are seemingly impossible to solve via classical VS approaches. We believe that this behavior is very difficult to model with analytic control approaches. On the other hand, the incorporation of the MoP auxiliary task (RPPO+IDM+MoP) allows the robot to improve its overall trajectories, reaching the desired pose with fewer maneuvers, and resulting in greater convergence accuracy.

Fig. 7(b) shows the trajectories performed by the robot on the test set after being trained respectively with our whole framework (RPPO+IDM+MoP), IBVS, IBVS-H [26], LVS [28], and VS-ViT [42]. As can be seen, in contrast to our approach, IBVS, IBVS-H [26], LVS [28], and VS-ViT collectively fail to overcome the nonholonomic constraints and converge to the desired pose. Notably, the final errors on the X_w -axis (Fig. 7(b)) for DRL-based methods are the highest among these methods, primarily due to the robot's limited ability to explore laterally, as discussed in Section 5.1. IBVS converges to a local optimum

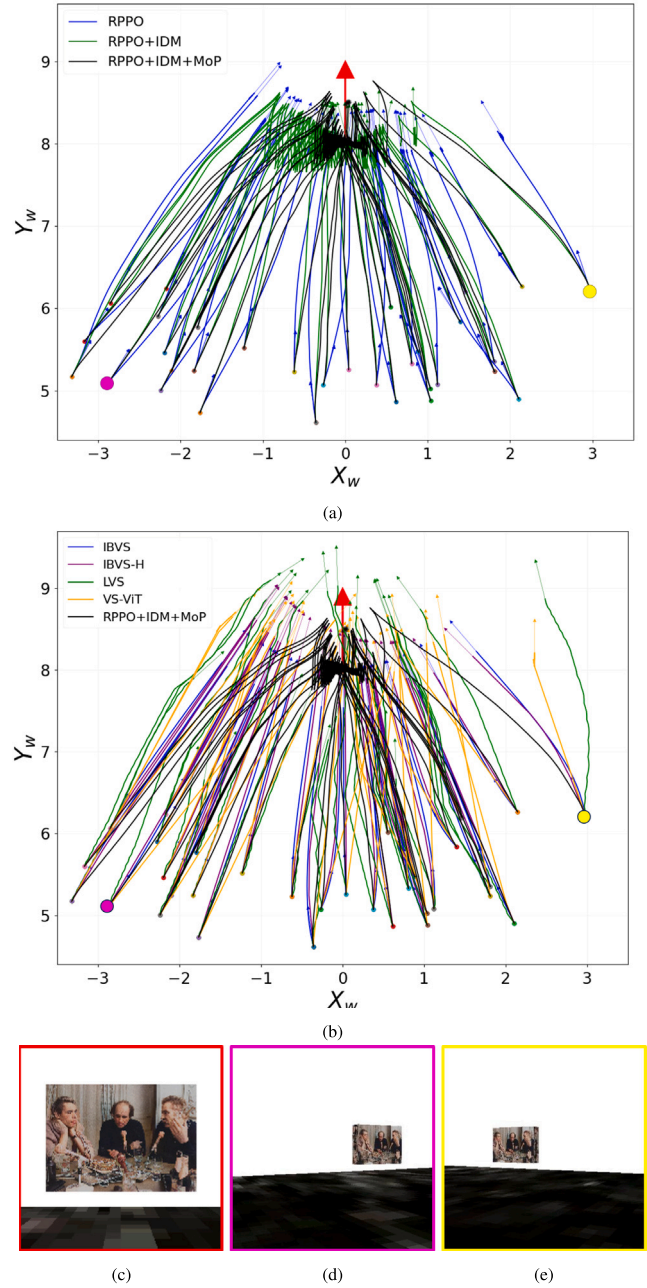


Fig. 7. Qualitative results: (a, b) Illustration of the trajectories performed by IBVS-H, VS-ViT, IBVS, and LVS compared to RPPO+IDM and RPPO+IDM+MoP, (c) The desired image acquired at the desired pose, shown by a red arrow, (d, e) Examples of initial images acquired at the poses, shown respectively by a pink and a yellow circles. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

due to the coupled visual degrees of freedom, which lead to a global optimum of the cost function in the feature space, despite the robot not reaching the desired pose. While the integration of DRL in IBVS-H and action clamping improves results compared to IBVS, IBVS-H is still unable to completely decouple its angular and linear velocities, thereby limiting its ability to perform complex trajectories. LVS fails to accurately achieve the desired pose due to the discrete action space and the lack of temporal representation. Although VS-ViT achieves better performance than LVS due to the improved representation provided by the ViT encoder, it still performs poorly because the agent in VS-ViT relies on the standard exploration mechanism in DRL, which restricts the set of trajectories discovered during training.

Table 2

Mean pose errors at convergence across our test set. ΔM is the average position error. $\Delta\beta$ is the average orientation error.

Method	ΔM (m)	$\Delta\beta$ (°)
Random policy	2.9898	21.7380
IBVS	0.8779	17.3663
IBVS-H	0.6926	19.0394
LVS	1.4420	18.8159
VS-ViT	0.8461	17.8705
RPPO	0.6905	18.7987
RPPO+IDM (ours)	0.2363	4.8186
RPPO+IDM+MoP (ours)	0.0552	2.1420
RPPO _o	0.0365	1.1173

In addition, we have quantitatively evaluated the performance of every method in terms of convergence accuracy. Table 2 provides the mean position and orientation errors at convergence across our test set. We note that the position error is the Euclidean distance between the desired position and the position of the robot at convergence. Conversely, the orientation error is the Euclidean distance between the desired orientation and the orientation of the robot at convergence. It is particularly interesting to observe that our approach (RPPO+IDM+MoP) can regulate the robot to the desired pose with a centimeter-level error, achieving accuracy similar to an omnidirectional robot (RPPO_o) (Table 2), and outperforming all compared methods. This result stems from the combination of the LSTM layer which allows to tackle the partial observability in vision-based control tasks, the intrinsic motivation model, which enables the robot to efficiently explore its environment and thus overcoming the limitations imposed by the nonholonomic constraints, and the auxiliary task, which enhances performance by providing the DRL model with a better representation. We suggest that the MoP auxiliary task may aid the agent in acquiring a representation of the interaction matrix. On the other hand, the choice of IDM in the intrinsic motivation model aims to learn features that depend on the robot's immediate controls to overcome the variability caused by visual domain randomization.

5.3. Comparison of intrinsic motivation models

We compare the intrinsic motivation model proposed in our framework to other existing approaches. Specifically, we combine the intrinsic curiosity models proposed in [65–67] with our base agent RPPO. The model proposed in [67], denoted as RPPO+RND, is based on rewarding the agent for predicting features of a random fixed CNN encoder. Similarly, the model presented in [65], denoted as RPPO+ICM, is based on rewarding the agent for predicting features of an IDM encoder. The model outlined in [66], denoted as RPPO+NGU, takes a different outlook on the problem by combining the method from [67] with an episodic memory to establish a count-based approach for assessing the novelty of visited states. To assess the efficiency of the IDM encoder, we train the method described in [60] with a fixed random CNN encoder, denoted as RPPO+CNN.

Fig. 8 shows the average rewards test curves of these trainings. A reminder that the objective of DRL is to maximize the expected discounted rewards. The higher the average test rewards, the better the performance. The pink bold line represents the average test rewards achieved by the omnidirectional robot, which corresponds to a robot capable of accurately positioning at the desired pose under no motion constraints. Our results show that using the approximation of Rényi divergence (RPPO+CNN and RPPO+IDM) for generating intrinsic rewards outperforms the compared methods. While it is not clear why this method is better, we assume that generating intrinsic rewards based on the approximation of Rényi divergence yields better results due to its reliance on observations from both the current and old policies, whereas the other methods rely only on the current policy. This bias, we believe, is what encourages the agent to try different

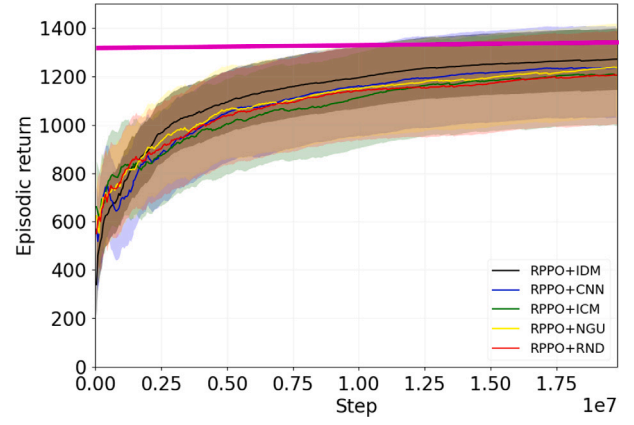


Fig. 8. Reward test curves for the different intrinsic motivation methods. The episodic return refers to the average rewards per episode. The bold pink line represents the average rewards achieved by the omnidirectional robot. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

trajectories as training progresses. A further performance improvement can be observed when using an IDM instead of a CNN encoder with the approach [60]. This is because the IDM allows the agent to rely only on the changes in its visual inputs that are impacted by its control velocities, and thus neglects the effects of domain randomization on input images.

5.4. Comparison of auxiliary tasks

To assess the efficiency of our MoP prediction module, we conduct comparisons with two other common auxiliary tasks: reward prediction (RP) and depth prediction (DP) introduced in [68,69]. The RP task consists of learning to predict rewards r_t from the observation o_t at each timestep. The DP task consists of predicting the depth map from the current image I_t . We use the method RPPO+IDM combined with the compared auxiliary tasks, namely RPPO+IDM+RP for the RP task and RPPO+IDM+DP for the DP task. Fig. 9 shows the average rewards test curves for these trainings. The DP task does not provide any improvement to RPPO+IDM. This is probably due to the geometric simplicity of our training environment (Fig. 4) that does not favor exploiting depth information. The RP task slightly improves the results due to its capacity to leverage information about the current reward

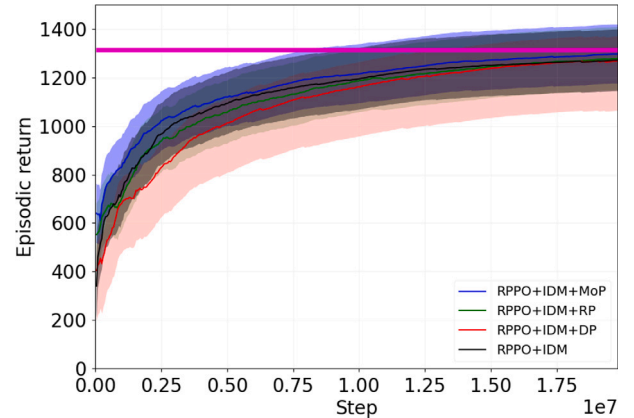


Fig. 9. Reward test curves for the compared auxiliary tasks. The bold pink line represents the average rewards achieved by the omnidirectional robot. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

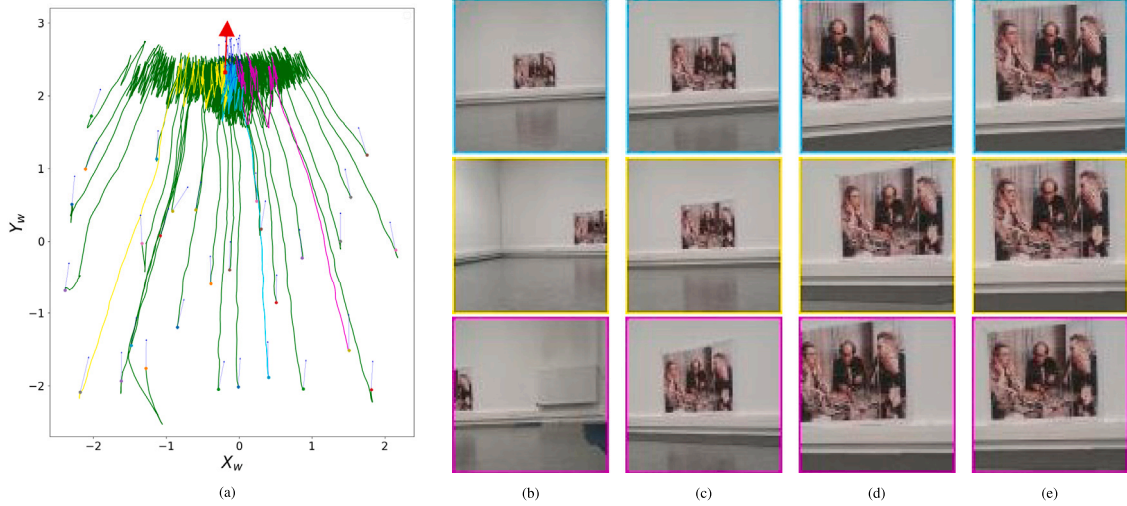


Fig. 10. Experiments conducted on the real robot: (a) the 30 trajectories performed by the real robot, (b) shows the initial images I_0 for the trajectories colored in pink (top), yellow (middle) and blue (bottom) respectively, (c, d) are transition images I_t acquired by the robot for each corresponding experiment, (e) illustrates the final images I_f acquired at convergence. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

into the agent's representation. On the other hand, our MoP auxiliary task results in higher rewards and thus, outperforms the compared tasks. We assume that by learning how control velocities impact the changes in the visual input, the agent is able to learn optimal control velocities in each state. These results show the importance of tailoring auxiliary tasks to the specific demands of the task at hand.

5.5. Generalization over desired poses

Generalization over desired poses refers to the capacity of the DRL model to drive the robot to any desired pose defined by an image. In this section, we aim to assess the generalization capacity of our method (RPPO+IDM+MoP) over a surface defined by a Gaussian distribution. In light of this, we conduct three experiments in which the desired pose is sampled during training from a 3D Gaussian distribution $N_3(\mu, \Sigma_i)$,

with $\mu = (x^*, y^*, \beta^*)$ and $\Sigma_i = \begin{bmatrix} \sigma_{x_i}^2 & 0 & 0 \\ 0 & \sigma_{y_i}^2 & 0 \\ 0 & 0 & \sigma_{\beta_i}^2 \end{bmatrix}$ is the covariance matrix

of the distribution. Here, $i = \{1, 2, 3\}$ is the index of the experiment. During training, the desired pose is randomly sampled at the beginning of each episode from its corresponding distribution. In these three experiments, $\beta^* = \arctan(\frac{x^*}{y^*})$ and $\sigma_{\beta_i} = 0.12$ is empirically fixed to ensure that the poster (Fig. 4(a)) is always, even partially, visible in the desired image regardless of the sampled desired orientation.

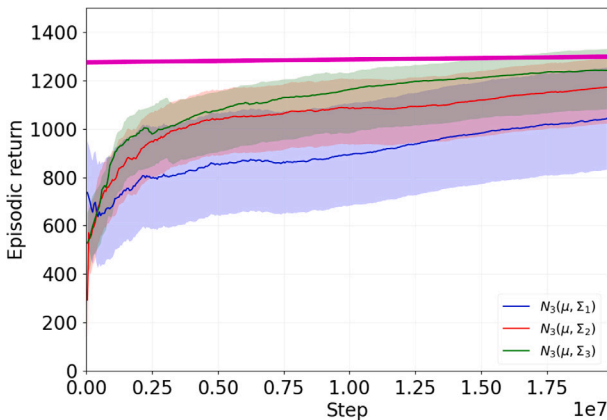


Fig. 11. Reward test curves for generalization experiments. The bold pink line represents the average rewards achieved by the omnidirectional robot.

Fig. 11 shows the average rewards test curves for the three experiments. Our results show that the larger the desired pose space is, the less the capacity of the model to generalize over that space. This result suggests that a DRL model is not able to fully capture a continuous representation that maps the desired images to the corresponding desired poses in the physical scene solely from a finite set of RGB images and rewards as a training signal. The problem of learning continuous visual representations for spatial understanding is a relatively new emerging subfield in robot learning [70,71].

5.6. Real-world transfer

Real experiments were conducted with a four-wheeled nonholonomic mobile robot (i.e., Robotnik Summit-XL) with a calibrated perspective camera mounted frontward (Fig. 4). It is important to note that the control policy was trained in simulation and deployed on the robot without any additional fine-tuning.

In order to qualitatively and quantitatively evaluate the robot's trajectories and its convergence accuracy, we used the direct lidar odometry (DLO) method [72] to continuously estimate the pose of the robot. More precisely, a static map of the scene has been first pre-built with DLO using the acquisitions of an embedded 3D Lidar. Once the map has been created, the robot has been manually controlled to the desired pose where the desired image has been acquired and saved, as well as, the desired pose thanks to DLO localization. It is worth noting that the map, the localization and the desired pose are only used for qualitative and quantitative evaluation and are not utilized by our model.

5.6.1. Validation and evaluation

The aim of this experiment is twofold: first, to validate the transfer of the learned control policy, and secondly, to evaluate the performance on a real robot using a set of 30 random initial poses that cover the entire room. Fig. 10 shows the trajectories obtained from these experiments. It can be seen that, similar to the simulation (Section 5.2), the robot is able to maneuver and recover from poses that are seemingly impossible to solve using VS or LVS.

Fig. 10(a) highlights three colored trajectories for which the initial images (Fig. 10(b)), images acquired during the robot's displacements (Figs. 10(c)–10(d)) and acquired at convergences (Fig. 10(e)) are shown. The mean position and orientation errors at convergence across these experiments are respectively $\Delta M = 0.2017$ m and $\Delta \beta = 7.9183^\circ$.

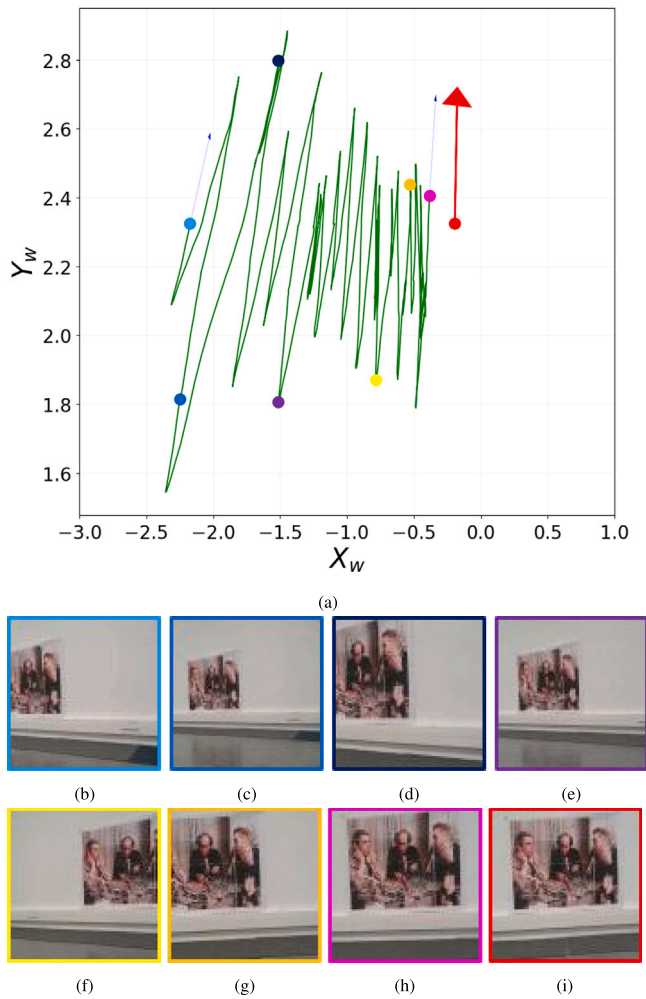


Fig. 12. Experiment performed on a real robot: (a) The trajectory executed by the robot, (b) The initial image captured at the initial pose, (c–h) sample images acquired along the trajectory for each corresponding, (i) The desired image.

Convergence precision in the real environment is slightly lower than the one obtained in simulation. This discrepancy is attributed to the variability in dynamics and visuals between simulation and real-world experiments. More specifically, friction with the ground influences maneuver performance, and the images rendered in simulation do not fully capture the details of the real scene. We emphasize that using visual domain randomization is crucial for transferring our control policy, as training the robot without randomization resulted in poor performance in the real scene. This is because adding noise to input images helps CNNs learn robust features, while domain randomization (Fig. 3) allows the agent to learn invariant features. Moreover, we noticed that, unlike in simulation, the robot can oscillate at convergence due to the nature of the stochastic policy.

5.6.2. Extreme case

The aim of this experiment is to study the robot's behavior in an arduous initial situation. Indeed, the robot is initially positioned approximately 2.5 m laterally from the desired pose with an angle of 5° . In such condition, the desired image is barely visible in the camera FOV (Fig. 12(b)) and, given that the difference in position is mainly on the robot's uncontrollable axis, classical, hybrid, or the existing learning-based VS approaches would have failed due to the FOV loss problem. In contrast, our DRL method enables the robot to autonomously maneuver in order to reach the desired pose (Fig. 12(a)), without losing the view from the poster during its displacement (Figs. 12(b)–12(i)).

6. Conclusion

In this paper, we present a deep reinforcement learning-based approach to address the challenge of visual control for nonholonomic mobile robots. We formulate the problem as a partially observable Markov decision process. Subsequently, we introduce a novel deep reinforcement learning method designed to effectively tackle the visual servoing task. Our approach incorporates a novel auxiliary task, which involves predicting motion perceptibility values from input images. Additionally, it includes an intrinsic motivation model designed to enhance the agent's exploration capabilities.

Our results highlight the challenging impact of nonholonomic constraints on the robot's ability to explore the environment. Furthermore, our approach outperforms the compared state of the art methods. It demonstrates an emergent control strategy, enabling the robot to recover from situations not achievable using traditional visual control schemes and effectively addressing the field of view loss problem. Additionally, we showcase the transferability of our model to a real robot.

However, our method has limitations, primarily due to the nature of DRL algorithms. Classical methods are plug-and-play and independent of specific environments, whereas DRL often requires training robots in simulation, which constitutes a bottleneck in terms of deployment on real robots.

Future work will focus on improving the transfer to the real world, as well as on training the model in a diverse set of environments under complex settings (including obstacles and field of view loss), utilizing more complex vision sensors, and understanding learning continuous representations for generalization across desired poses.

CRedit authorship contribution statement

Takieddine Soualhi: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Nathan Crombez:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **Alexandre Lombard:** Writing – review & editing, Validation, Supervision. **Yassine Ruichek:** Writing – review & editing, Validation, Supervision, Conceptualization. **Stéphane Galland:** Writing – review & editing, Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Complexity analysis

In this section, we present a complexity analysis of our proposed method, discussing both computational and time complexities for training and testing. Table A.3 illustrated details of the architecture of the proposed method. We note that for the architectural details, we follow the PyTorch documentation convention.

Training: We report the training complexity in terms of processed frames per second (FPS) and GPU usage (reported as a percentage), measured on a single NVIDIA RTX 3090 GPU with 24 GB of memory. This measurement includes both forward and backward passes, as well as the overhead from running simulations in 48 parallel PyBullet environments. The computational complexity is primarily influenced by the MoP auxiliary task, which led to a noticeable decrease in training frame rates when incorporating this task. Nonetheless, the training process still maintains high frame rates overall (see Table A.4).

Table A.3

Details of the architecture of the proposed method.

Model	Layer	Architecture
Policy model	Conv2D ₁	(3,32,8)
	Conv2D ₂	(32,64,4)
	Conv2D ₃	(64,64 3)
	FC ₁	(2304,256)
	LSTM	(515,1024)
	FC ₂ (V_f)	(1024,1024)
	FC ₃ (π)	(1024,1024)
Inverse dynamics model	FC ₄ (MoP)	(1024,1024)
	Conv2D ₄	(3,32,8)
	Conv2D ₅	(32,64,4)
	Conv2D ₆	(64,64 3)
	FC ₅	(1152,128)

Table A.4

Complexity analysis.

Method	FPS	GPU usage (%)
RPPO	105	65
RPPO+IDM	105	86
RPPO+IDM+MoP	90	92

Testing: During inference on the real robot, only a forward pass through the policy model is required. The GPU on the real robot is an NVIDIA RTX 2080 Ti with 12 GB of memory. The policy model alone, when loaded, requires around 2 GB of memory, which corresponds to 16% of the available memory. We record around a 3 ms response time, demonstrating the system's ability to meet the timing requirements of real-time robotics. These results confirm the architecture's reliability and efficiency, as validated by our real robot experiments.

Appendix B. Supplementary data

A video of the simulation and real-world experiments is available at: <https://youtu.be/2srTt8YQwCs>.

Data availability

Data will be made available on request.

References

- [1] J.-P. Laumond, S. Sekhavat, F. Lamiraux, Guidelines in nonholonomic motion planning for mobile robots, *Robot. Motion Plan. Control.* (2005) 1–53.
- [2] F. Conticelli, B. Allotta, P.K. Khosla, Image-based visual servoing of nonholonomic mobile robots, in: *Proceedings of the 38th IEEE Conference on Decision and Control*, 1999, pp. 3496–3501.
- [3] F. Chaumette, S. Hutchinson, Visual servo control. I. Basic approaches, *IEEE Robot. Autom. Mag.* (2006) 82–90.
- [4] N.R. Gans, S.A. Hutchinson, A stable vision-based control scheme for nonholonomic vehicles to keep a landmark in the field of view, in: *IEEE International Conference on Robotics and Automation*, 2007, pp. 2196–2201.
- [5] Y. Masutani, M. Mikawa, N. Maru, F. Miyazaki, Visual servoing for nonholonomic mobile robots, in: *Proceedings of IEEE International Conference on Intelligent Robots and Systems*, 1994, pp. 1133–1140.
- [6] X. Zhang, Y. Fang, X. Liu, Motion-estimation-based visual servoing of nonholonomic mobile robots, *IEEE Trans. Robot.* (2011) 1167–1175.
- [7] H. Aliakbarpour, O. Tahri, H. Araujo, Visual servoing of mobile robots using non-central catadioptric cameras, *Robot. Auton. Syst.* (2014) 1613–1622.
- [8] Y. Alj, G. Caron, Featureless omnidirectional vision-based control of nonholonomic mobile robot, in: *IEEE International Conference on Ubiquitous Robots and Ambient Intelligence*, 2015, pp. 95–100.
- [9] B.R. Kiran, I. Sobh, V. Talpaert, P. Mannion, A.A. Al Sallab, S. Yogamani, P. Pérez, Deep reinforcement learning for autonomous driving: A survey, *IEEE Trans. Intell. Transp. Syst.* (2021) 4909–4926.
- [10] H. Wang, W. Wang, X. Zhu, J. Dai, L. Wang, Collaborative visual navigation, 2021, arXiv preprint arXiv:2107.01151.
- [11] S. Joshi, S. Kumra, F. Sahin, Robotic grasping using deep reinforcement learning, in: *IEEE International Conference on Automation Science and Engineering*, 2020, pp. 1461–1466.
- [12] R.T. Fomena, F. Chaumette, Improvements on visual servoing from spherical targets using a spherical projection model, *IEEE Trans. Robot.* (2009) 874–886.
- [13] X. Liang, H. Wang, W. Chen, Adaptive image-based visual servoing of wheeled mobile robots with fixed camera configuration, in: *IEEE International Conference on Robotics and Automation*, ICRA, IEEE, 2014, pp. 6199–6204.
- [14] X. Zhang, Y. Fang, X. Liu, Motion-estimation-based visual servoing of nonholonomic mobile robots, *IEEE Trans. Robot.* 27 (6) (2011) 1167–1175.
- [15] G. Chesi, K. Hashimoto, D. Prattichizzo, A. Vicino, Keeping features in the field of view in eye-in-hand visual servoing: a switching approach, *IEEE Trans. Robot.* 20 (5) (2004) 908–914.
- [16] F. Ke, Z. Li, H. Xiao, X. Zhang, Visual servoing of constrained mobile robots based on model predictive control, *IEEE Trans. Syst. Man Cybern.: Syst.* 47 (7) (2017) 1428–1438.
- [17] Y. Wang, H. Lang, C.W. De Silva, A hybrid visual servo controller for robust grasping by wheeled mobile robots, *Trans. Mechatronics* (2009).
- [18] G. Allibert, E. Courtial, F. Chaumette, Predictive control for constrained image-based visual servoing, *IEEE Trans. Robot.* 26 (5) (2010) 933–939.
- [19] S. Heshmati-Alamdari, G.K. Karavas, A. Eqtami, M. Drossakis, K.J. Kyriakopoulos, Robustness analysis of model predictive control for constrained image-based visual servoing, in: *2014 IEEE International Conference on Robotics and Automation*, ICRA, IEEE, 2014, pp. 4469–4474.
- [20] S. Heshmati-Alamdari, G.C. Karras, A. Eqtami, K.J. Kyriakopoulos, A robust self triggered image based visual servoing model predictive control scheme for small autonomous robots, in: *International Conference on Intelligent Robots and Systems*, IROS, IEEE, 2015.
- [21] S. Hutchinson, P. Corke, A new partitioned approach to image-based visual servo control, in: *International Symposium on Robotics*, IEEE, 2000.
- [22] A. Remazeilles, N. Mansard, F. Chaumette, A qualitative visual servoing to ensure the visibility constraint, in: *2006 IEEE/RSJ International Conference on Intelligent Robots and Systems*, IEEE, 2006.
- [23] Z. Miljković, M. Mitić, M. Lazarević, B. Babić, Neural network reinforcement learning for visual control of robot manipulators, *Expert Syst. Appl.* (2013).
- [24] H. Shi, X. Li, K.-S. Hwang, W. Pan, G. Xu, Decoupled visual servoing with fuzzy Q-learning, *Trans. Ind. Inform.* (2016).
- [25] H. Shi, L. Shi, G. Sun, K.-S. Hwang, Adaptive image-based visual servoing for hovering control of quad-rotor, *Trans. Cogn. Dev. Syst.* (2019).
- [26] Z. Jin, J. Wu, A. Liu, W.-A. Zhang, L. Yu, Policy-based deep reinforcement learning for visual servoing control of mobile robots with visibility constraints, *IEEE Trans. Ind. Electron.* 69 (2) (2021) 1898–1908.
- [27] J. Wu, Z. Jin, A. Liu, L. Yu, F. Yang, A hybrid deep-q-network and model predictive control for point stabilization of visual servoing systems, *Control Eng. Pract.* 128 (2022) 105314.
- [28] Y. Li, J. Košćeka, Learning view and target invariant visual servoing for navigation, in: *IEEE International Conference on Robotics and Automation*, 2020, pp. 658–664.
- [29] Y. Choi, S. Oh, Image-goal navigation via keypoint-based reinforcement learning, in: *IEEE International Conference on Ubiquitous Robots*, 2021, pp. 18–21.
- [30] S. Levine, C. Finn, T. Darrell, P. Abbeel, End-to-end training of deep visuomotor policies, *J. Mach. Learn. Res.* (2016) 1334–1373.
- [31] F. Sadeghi, A. Toshev, E. Jang, S. Levine, Sim2real view invariant visual servoing by recurrent control, 2017, arXiv preprint arXiv:1712.07642.
- [32] S. Asayesh, H.S. Darani, M. Mehrandezh, K. Gupta, et al., Toward scalable visual servoing using deep reinforcement learning and optimal control, 2023, arXiv preprint arXiv:2310.01360.
- [33] E.G. Ribeiro, R. de Queiroz Mendes, V. Grassi Jr., Real-time deep learning approach to visual servo control and grasp detection for autonomous robotic manipulation, *Robot. Auton. Syst.* 139 (2021) 103757.
- [34] O.-M. Pedersen, E. Misimi, F. Chaumette, Grasping unknown objects by coupling deep reinforcement learning, generative adversarial networks, and visual servoing, in: *IEEE International Conference on Robotics and Automation*, 2020, pp. 5655–5662.
- [35] Y. Zhu, R. Mottaghi, E. Kolve, J.J. Lim, A. Gupta, L. Fei-Fei, A. Farhadi, Target-driven visual navigation in indoor scenes using deep reinforcement learning, in: *IEEE International Conference on Robotics and Automation*, 2017, pp. 3357–3364.
- [36] J. Kulháněk, E. Derner, T. De Bruin, R. Babuška, Vision-based navigation using deep reinforcement learning, in: *European Conference on Mobile Robots*, 2019, pp. 1–8.
- [37] J. Ho, S. Ermon, Generative adversarial imitation learning, *Adv. Neural Inf. Process. Syst.* 29 (2016).
- [38] Q. Wu, X. Gong, K. Xu, D. Manocha, J. Dong, J. Wang, Towards target-driven visual navigation in indoor scenes via generative imitation learning, *IEEE Robot. Autom. Lett.* (2020) 175–182.
- [39] F. Sadeghi, Divis: Domain invariant visual servoing for collision-free goal reaching, 2019, arXiv preprint arXiv:1902.05947.
- [40] L. Mezghani, S. Sukhbaatar, T. Lavril, O. Maksymets, D. Batra, P. Bojanowski, K. Alahari, Memory-augmented reinforcement learning for image-goal navigation, 2021, arXiv preprint arXiv:2101.05181.
- [41] A. Majumdar, G. Aggarwal, B. Devnani, J. Hoffman, D. Batra, Zson: Zero-shot object-goal navigation using multimodal goal embeddings, *Adv. Neural Inf. Process. Syst.* 35 (2022) 32340–32352.

- [42] K. Yadav, A. Majumdar, R. Ramrakhya, N. Yokoyama, A. Baevski, Z. Kira, O. Maksymets, D. Batra, Ovrl-v2: A simple state-of-art baseline for imagenav and objectnav, 2023, arXiv preprint [arXiv:2303.07798](https://arxiv.org/abs/2303.07798).
- [43] K. Yadav, R. Ramrakhya, A. Majumdar, V.-P. Berges, S. Kuhar, D. Batra, A. Baevski, O. Maksymets, Offline visual representation learning for embodied navigation, in: *Workshop on Reincarnating Reinforcement Learning At ICLR* 2023, 2023.
- [44] A. Majumdar, K. Yadav, S. Arnaud, Y.J. Ma, C. Chen, S. Silwal, A. Jain, V.-P. Berges, P. Abbeel, J. Malik, D. Batra, Y. Lin, O. Maksymets, A. Rajeswaran, F. Meier, Where are we in the search for an artificial visual cortex for embodied intelligence? 2023, [arXiv:2303.18240](https://arxiv.org/abs/2303.18240).
- [45] S.K. Ramakrishnan, A. Gokaslan, E. Wijmans, O. Maksymets, A. Clegg, J. Turner, E. Undersander, W. Galuba, A. Westbury, A.X. Chang, et al., Habitat-matterport 3d dataset (hm3d): 1000 large-scale 3d environments for embodied ai, 2021, arXiv preprint [arXiv:2109.08238](https://arxiv.org/abs/2109.08238).
- [46] W. Luo, P. Sun, F. Zhong, W. Liu, T. Zhang, Y. Wang, End-to-end active object tracking and its real-world deployment via reinforcement learning, *IEEE Trans. Pattern Anal. Mach. Intell.* 42 (6) (2019) 1317–1332.
- [47] F. Zhong, P. Sun, W. Luo, T. Yan, Y. Wang, Ad-vat+: An asymmetric dueling mechanism for learning and understanding visual active tracking, *IEEE Trans. Pattern Anal. Mach. Intell.* 43 (5) (2019) 1467–1482.
- [48] A. Devo, A. Dionigi, G. Costante, Enhancing continuous control of mobile robots for end-to-end visual active tracking, *Robot. Auton. Syst.* 142 (2021) 103799.
- [49] A. Dionigi, A. Devo, L. Guiducci, G. Costante, E-vat: An asymmetric end-to-end approach to visual active exploration and tracking, *IEEE Robot. Autom. Lett.* 7 (2) (2022) 4259–4266.
- [50] K. Tchoń, J. Ratajczak, Singularities of holonomic and non-holonomic robotic systems: A normal form approach, *J. Franklin Inst.* 358 (15) (2021) 7698–7713.
- [51] A. Aubret, L. Matignon, S. Hassas, A survey on intrinsic motivation in reinforcement learning, 2019, arXiv [abs/1908.06976](https://arxiv.org/abs/1908.06976).
- [52] R.S. Sutton, A.G. Barto, *Reinforcement Learning: An Introduction*, MIT Press, 2018.
- [53] O.M. Andrychowicz, B. Baker, M. Chociej, R. Jozefowicz, B. McGrew, J. Pachocki, A. Petron, M. Plappert, G. Powell, A. Ray, et al., Learning dexterous in-hand manipulation, *Int. J. Robot. Res.* (2020) 3–20.
- [54] S. Hochreiter, J. Schmidhuber, Long short-term memory, *Neural Comput.* (1997) 1735–1780.
- [55] M. Hausknecht, P. Stone, Deep recurrent q-learning for partially observable mdps, in: *AAAI Fall Symposium Series*, 2015.
- [56] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, O. Klimov, Proximal policy optimization algorithms, 2017, arXiv preprint [arXiv:1707.06347](https://arxiv.org/abs/1707.06347).
- [57] C. Lyle, M. Rowland, G. Ostrovski, W. Dabney, On the effect of auxiliary tasks on representation dynamics, in: *International Conference on Artificial Intelligence and Statistics*, PMLR, 2021, pp. 1–9.
- [58] R. Sharma, S. Hutchinson, Motion perceptibility and its application to active vision-based servo control, *IEEE Trans. Robot. Autom.* (1997) 607–617.
- [59] C. Collewet, E. Marchand, Photometric visual servoing, *IEEE Trans. Robot.* (2011) 828–834.
- [60] M. Yuan, B. Li, X. Jin, W. Zeng, Rewarding episodic visitation discrepancy for exploration in reinforcement learning, 2022, arXiv preprint [arXiv:2209.08842](https://arxiv.org/abs/2209.08842).
- [61] Y. Burda, H. Edwards, D. Pathak, A. Storkey, T. Darrell, A.A. Efros, Large-scale study of curiosity-driven learning, 2018, arXiv preprint [arXiv:1808.04355](https://arxiv.org/abs/1808.04355).
- [62] E. Coumans, Y. Bai, Pybullet, a python module for physics simulation for games, robotics and machine learning, 2016.
- [63] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, N. Dormann, Stable-baselines3: Reliable reinforcement learning implementations, *J. Mach. Learn. Res.* (2021) 12348–12355.
- [64] T. Akiba, S. Sano, T. Yanase, T. Ohta, M. Koyama, Optuna: A next-generation hyperparameter optimization framework, in: *ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2019, pp. 2623–2631.
- [65] D. Pathak, P. Agrawal, A.A. Efros, T. Darrell, Curiosity-driven exploration by self-supervised prediction, in: *International Conference on Machine Learning*, PMLR, 2017, pp. 2778–2787.
- [66] A.P. Badia, P. Sprechmann, A. Vitvitskiy, D. Guo, B. Piot, S. Kapturowski, O. Tieleman, M. Arjovsky, A. Pritzel, A. Bolt, et al., Never give up: Learning directed exploration strategies, 2020, arXiv preprint [arXiv:2002.06038](https://arxiv.org/abs/2002.06038).
- [67] Y. Burda, H. Edwards, A. Storkey, O. Klimov, Exploration by random network distillation, 2018, arXiv preprint [arXiv:1810.12894](https://arxiv.org/abs/1810.12894).
- [68] M. Jaderberg, V. Mnih, W.M. Czarnecki, T. Schaul, J.Z. Leibo, D. Silver, K. Kavukcuoglu, Reinforcement learning with unsupervised auxiliary tasks, 2016, arXiv preprint [arXiv:1611.05397](https://arxiv.org/abs/1611.05397).
- [69] P. Mirowski, R. Pascanu, F. Viola, H. Soyer, A.J. Ballard, A. Banino, M. Denil, R. Goroshin, L. Sifre, K. Kavukcuoglu, et al., Learning to navigate in complex environments, 2016, arXiv preprint [arXiv:1611.03673](https://arxiv.org/abs/1611.03673).

- [70] Y. Li, S. Li, V. Sitzmann, P. Agrawal, A. Torralba, 3D neural scene representations for visuomotor control, in: *Conference on Robot Learning*, PMLR, 2022, pp. 112–123.
- [71] D. Driess, I. Schubert, P. Florence, Y. Li, M. Toussaint, Reinforcement learning with neural radiance fields, *Adv. Neural Inf. Process. Syst.* (2022) 16931–16945.
- [72] K. Chen, B.T. Lopez, A.-a. Agha-mohammadi, A. Mehta, Direct LiDAR odometry: Fast localization with dense point clouds, *IEEE Robot. Autom. Lett.* (2022) 2000–2007.



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